

Comparative Analysis of Sampling Strategies and Scalable Clustering Algorithms for Emotion Discovery on Imbalanced Text Data

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Abstract—This paper investigates discovering natural emotion clusters in text data using unsupervised learning methods. The dataset contains 473,247 instances with 7 emotion labels (joy, sadness, anger, love, fear, surprise, disgust), with significant class imbalance (disgust 3.97%, joy 30.32%). Text representation uses TF-IDF and Word2Vec embeddings. Six methods are compared: four K-means variants with different sampling strategies, K-means on Word2Vec, and DBSCAN on UMAP-reduced TF-IDF. Results show that all K-means methods achieve very low ARI values (-0.0056 to 0.0231), indicating TF-IDF and K-means cannot discover meaningful emotion clusters. Word2Vec produces more balanced clusters but does not improve quality (ARI = 0.0199). DBSCAN achieves the highest NMI (0.1334) and purity (87.12%), but with extreme fragmentation (226 clusters) and negative Silhouette (-0.0622). It is concluded that emotion expressions form many small, dense regions rather than well-separated groups, and that labeled emotion categories do not align with natural structures in the data.

Abstract—Ovaj rad istražuje otkrivanje prirodnih emocionalnih klastera u tekstualnim podacima korištenjem metoda nenadziranog učenja. Skup podataka sadrži 473.247 instanci sa 7 emocionalnih labela (radost, tuga, ljutnja, ljubav, strah, iznenađenje, gađenje), sa značajnom klasnom neravnotežom (gađenje 3.97%, radost 30.32%). Za predstavljanje teksta koriste se TF-IDF i Word2Vec vektorske reprezentacije. U radu se poredi 6 metoda: 4 K-means varijante s različitim strategijama uzorkovanja, K-means na Word2Vec reprezentacijama i DBSCAN na UMAP-reduciranim TF-IDF karakteristikama. Rezultati pokazuju da sve K-means metode postižu izuzetno niske ARI vrijednosti (-0.0056 do 0.0231), što ukazuje da TF-IDF i K-means ne mogu otkriti smislene emocionalne klastere. Word2Vec daje uravnoteženije klastere ali ne poboljšava kvalitet (ARI = 0.0199). DBSCAN postiže najveću NMI (0.1334) i čistoću (87.12%), ali uz ekstremnu fragmentaciju sa 226 klastera i negativnu Silhouette ocjenu (-0.0622). Zaključuje se da emocionalni izrazi formiraju mnogo malih, gustih regija umjesto nekoliko dobro odvojenih grupa, te da se labelirani emocionalni koncepti ne podudaraju s prirodnim strukturama u podacima.

Index Terms—K-means clustering, DBSCAN, Word2Vec, TF-IDF, UMAP, emotion recognition, unsupervised learning, class imbalance, text clustering

I. INTRODUCTION

Emotion recognition from text is a fundamental task in natural language processing (NLP)[1] with wide-ranging applications in mental health monitoring, au-

tomatic opinion mining, emotion-aware conversational agents, therapeutic practices [2], healthcare, education, and autonomous systems [3]. While supervised approaches have achieved considerable success in detecting predefined emotion categories, they rely on large amounts of labeled data and are constrained by the specific taxonomies used during training [4]. In contrast, unsupervised learning offers the potential to discover natural emotional structures that may not align with human-defined categories [5], [6], revealing how emotions are actually expressed in real-world text [7]. This is particularly valuable in domains where emotion labels are ambiguous, culturally dependent, or continuously evolving, such as in social media discourse where new expressive forms emerge rapidly [8].

A significant body of work has explored text clustering for topic discovery and document organization using methods such as K-means [9] with TF-IDF features [4]. However, applying these techniques to emotion detection presents unique challenges. Emotions are expressed through diverse vocabulary and nuanced language, making semantic similarity difficult to capture with word-frequency-based representations. For instance, the expressions "I feel down" and "I'm feeling blue" convey the same emotional state but share no overlapping vocabulary. Furthermore, real-world emotion datasets are often imbalanced, with some emotions occurring far more frequently than others, which can bias clustering algorithms toward majority classes. Recent work on emotion datasets, such as GoEmotions [10], has highlighted the complexity of emotion annotation and the need for methods that can handle fine-grained emotional distinctions across diverse text genres.

To address the limitations of word-frequency-based representations, researchers have turned to dense vector representations such as Word2Vec [11] and more recently transformer-based embeddings like BERT [12]. These methods capture semantic relationships between words by learning from large corpora, enabling more meaningful similarity measurements between texts that express the same emotion using different vocabulary. In the context of clustering, Mikolov et al. (2013)

demonstrated that continuous vector representations can capture linguistic regularities and semantic relationships, making them suitable for downstream tasks such as document clustering. Additionally, clustering algorithms that do not assume spherical cluster shapes, such as DBSCAN [13], have been proposed to handle the complex, non-spherical distributions often found in high-dimensional text data. DBSCAN defines clusters as dense regions separated by regions of lower density, making it capable of discovering clusters of arbitrary shape and identifying noise points that do not belong to any cluster. Dimensionality reduction techniques like UMAP [14] further enable the application of density-based clustering methods by mitigating the curse of dimensionality while preserving the local and global structure of the data.

This paper investigates the problem of discovering natural emotion clusters in a large-scale text dataset using unsupervised learning methods. These are the six systematically compared approaches: (A) standard K-means on a simple random sample, (B) repeated K-means for stability analysis, (C) stratified K-means, (D) purposefully balanced K-means, (E) K-means on Word2Vec sentence embeddings to assess the impact of semantic representations, and (F) DBSCAN on UMAP-reduced TF-IDF features to test whether non-spherical clusters better capture emotion structure. Through this comprehensive comparison, this paper aims to identify the key factors that enable or hinder effective emotion clustering and provide insights for future work in this domain.

The dataset¹² used in this work contains 473,250 text instances distributed across 7 emotion labels with the class distribution shown in Table I.

TABLE I: Class Distribution of the Emotion Dataset

Emotion Label	Count	Percentage	Cumulative
Joy	143,067	30.2307%	30.23%
Sadness	121,187	25.6074%	55.84%
Anger	59,317	12.5340%	68.37%
Love	57,356	12.1196%	80.49%
Fear	49,649	10.4911%	90.98%
Surprise	22,640	4.7839%	95.77%
Disgust	20,034	4.2333%	100.00%
Total	473,250		

The chosen dataset exhibits severe class imbalance, with joy being the majority class comprising 30.3% of instances and disgust the minority class comprising mere 3.97%. This imbalance poses a big challenge for unsupervised clustering since algorithms may fail to discover clusters corresponding to under-represented emotions.

¹Data are consisted of a few datasets (1 full from Kaggle; then 7 sampled datasets: 6 from Kaggle and SemEval-2018; 1 custom synthetic dataset, and the augmented (paraphrased + backtranslated) disgust instances)

²Full data guide: <https://github.com/kanitafro/emotions-data-kmeans/tree/main/data>

The seven emotions—anger, love, joy, surprise, sadness, fear, and disgust—are well-justified for text-based emotion detection. This set draws from foundational psychological models, including Ekman’s six basic emotions [15], Parrot’s model [16], Russell’s circumplex [17], and Plutchik’s wheel [18], which converge on these as primary affective states. Empirically, these emotions are widely adopted in text-based emotion detection, with successful models achieving high performance, such as Chiorrini et al. (89% F1) [19] and recent works exceeding 90% accuracy [20], [21]. Their prevalence in standard datasets like SemEval-2007 [22], ISEAR [23], and Fairy Tales [24] ensures replicability and cross-study comparison. This theoretical grounding and empirical success confirm these emotions as a reliable choice for robust emotion detection systems.

II. USED ALGORITHM

A. K-Means Clustering

K-means [9] is a centroid-based partitioning algorithm that aims to partition n observations into k clusters, where each observation belongs to the cluster with the nearest mean (centroid). The algorithm minimizes the within-cluster sum of squares (WCSS), also known as inertia:

$$\text{Inertia} = \sum_{i=1}^k \sum_{\mathbf{x} \in C_i} \|\mathbf{x} - \boldsymbol{\mu}_i\|^2 \quad (1)$$

where $\boldsymbol{\mu}_i$ is the centroid of cluster C_i .

The algorithm proceeds iteratively:

- 1) **Initialization:** Select k initial centroids using K-means++ [25] for improved initialization.
- 2) **Assignment:** Assign each data point to the nearest centroid using Euclidean distance.
- 3) **Update:** Recalculate each centroid as the mean of all points assigned to it.
- 4) **Convergence:** Repeat steps 2-3 until centroids stabilize.

This paper uses K-means clustering algorithm as implemented in scikit-learn and as such it employs the K-means++ initialization method. K-means++ spreads initial centroids by sampling points with probability proportional to their squared distance from already chosen centroids, which accelerates convergence and improves cluster quality compared to purely random initialization [25].

B. TF-IDF Vectorization

For text data, documents are first converted to TF-IDF (Term Frequency-Inverse Document Frequency) [26], [27] feature vectors. The TF-IDF weight for term t in document d is computed as:

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \log \left(\frac{N}{\text{DF}(t)} \right) \quad (2)$$

where $\text{TF}(t, d)$ is the frequency of term t in document d , N is the total number of documents, and

$DF(t)$ is the number of documents containing term t . This representation emphasizes terms that are frequent in a specific document but rare across the entire corpus, capturing the distinctive vocabulary of each document.

C. Word2Vec Embeddings

Word2Vec [11] is a neural network-based technique for learning dense vector representations of words from large text corpora. The model learns to predict a word from its context (Continuous Bag-of-Words) or to predict context words from a target word (Skip-gram), producing embeddings where semantically similar words are located close to each other in the vector space. For sentence-level representation, Word2Vec embeddings of all words in the text are averaged producing a 300-dimensional dense vector that captures the overall semantic content of the sentence. This approach addresses the limitations of TF-IDF by capturing semantic similarity between words that do not co-occur—for example, *"I feel down"* and *"I'm feeling blue"* have similar embeddings despite sharing no overlapping vocabulary.

D. DBSCAN

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) [13] is a density-based clustering algorithm that does not assume spherical cluster shapes. It defines clusters as dense regions separated by regions of lower density, making it capable of discovering clusters of arbitrary shape. The algorithm relies on two parameters: ϵ , the radius of a neighborhood, and $minPts$, the minimum number of points required to form a dense region. Points that belong to no dense region are classified as noise. This makes DBSCAN particularly suitable for text data, where emotion expressions form highly overlapping, non-spherical distributions that K-means cannot effectively separate.

E. UMAP

UMAP (Uniform Manifold Approximation and Projection) [14] is a manifold learning technique for dimensionality reduction. It constructs a high-dimensional graph representation of the data and then optimizes a low-dimensional embedding that preserves the topological structure of the original space. Unlike linear methods such as PCA, UMAP captures non-linear relationships in the data and is among the fastest manifold learning algorithms. In this work, UMAP is used here to reduce the 5,000-dimensional TF-IDF feature space to 50 dimensions, enabling the application of DBSCAN while mitigating the curse of dimensionality.

F. Sampling Strategies for K-Means

1) *Method A: Single Random Sampling (Baseline)*: A single random sample of 21,000 instances is drawn from the full 473,247-instance dataset using simple random sampling. This method serves as the baseline for evaluating whether sampling strategies affect

cluster discovery. While simple and fast, its results depend on which 21,000 samples are selected, and minority classes may be severely under-represented (disgust would appear only approximately 830 times in a 21,000 sample).

2) *Method B: Repeated Sampling (Stability Analysis)*: Ten independent 21,000-instance random samples are processed with K-means to assess result stability. This approach, inspired by bootstrap methods in clustering validation [4], provides statistical confidence intervals for the optimal number of clusters and demonstrates that discovered structures are not artifacts of sampling variation. The optimal k is determined for each iteration, and the mode across iterations is reported.

3) *Method C: Stratified Sampling (Natural Distribution)*: Stratified sampling preserves the original class distribution of the full dataset in the 21,000-instance sample. For each emotion label l , the number of samples drawn is proportional to its percentage in the full dataset:

$$n_l = \text{round} \left(21000 \times \frac{\text{count}_l}{473247} \right) \quad (3)$$

This method answers the question: *What are the natural cluster structures in the data as it exists in the wild?*

4) *Method D: Purposefully Balanced Sampling (Scalability Test)*: Exactly 3,000 samples are drawn from each of the 7 emotion labels, resulting in a 21,000-instance training set. This approach removes the imbalance confounder entirely, allowing us to answer: *Can the algorithm distinguish between all 7 emotions if given equal opportunity, and would scaling to the full dataset help?*

The viability of scaling is tested through three complementary analyses:

- 1) **Centroid stability**: Cluster centers are compared across multiple balanced samples. If centroids are consistent across runs, the model is robust.
- 2) **Learning curves**: Performance is measured with increasing samples per label (500, 1000, 1500, 2000, 2500, 3000). Diminishing returns indicate that 3,000 samples are sufficient.
- 3) **Out-of-sample validation**: The model trained on 21,000 balanced samples is evaluated on 10,000 unseen samples from the full dataset. A small train-test gap (<0.05 ARI) indicates strong generalization.

5) *Method E: Word2Vec + K-Means (Semantic Sampling)*: Method E draws a single random sample of 21,000 instances from the full dataset, identical to Method A's sampling strategy. However, instead of TF-IDF vectorization, the text is represented using Word2Vec sentence embeddings. This approach tests whether semantic representations improve clustering quality compared to word-frequency-based features, while keeping the sampling strategy constant to isolate the effect of the representation change.

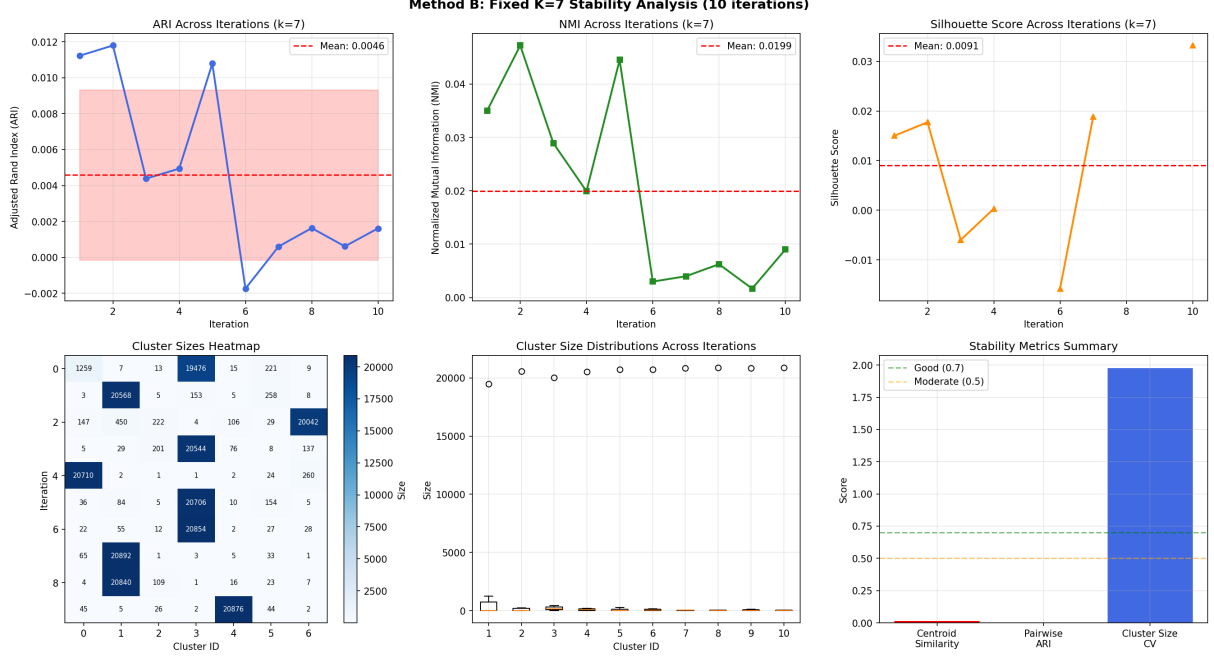


Fig. 1: Stability analysis for Method B with a fixed $k = 7$ across 10 iterations. Top row shows fluctuations in ARI, NMI, and Silhouette scores that shows poor cluster quality and near-random agreement. Bottom row displays severe cluster size imbalance, where a single random cluster captures 90–99% of the data in each iteration, resulting in an extremely high Cluster Size Coefficient of Variation (CV).

6) *Method F: DBSCAN + TF-IDF + UMAP (Density-Based Sampling)*: Method F draws a single random sample of 21,000 instances from the full dataset, consistent with Methods A and E. The key difference lies in the clustering algorithm and preprocessing: TF-IDF features are reduced to 50 dimensions using UMAP, and DBSCAN is applied instead of K-means. This approach tests whether density-based clustering on reduced-dimensionality TF-IDF features can better capture non-spherical emotion clusters. The random sampling strategy ensures comparability with Methods A and E, while the algorithm and preprocessing changes isolate the effects of density-based clustering and dimensionality reduction.

G. Evaluation Metrics

a) *Silhouette Score*: This metric [28] measures how similar a point is to its own cluster compared to other clusters. For a point i assigned to cluster A , let $a(i)$ be the mean distance to other points in A , and $b(i)$ be the minimum mean distance to points in any other cluster. The silhouette score is:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \quad (4)$$

The score ranges from -1 (poor clustering) to $+1$ (excellent clustering). Mean silhouette score are reported across all points.

b) *Adjusted Rand Index (ARI)*: This metric [29] measures the agreement between two clusterings (e.g., predicted clusters vs ground truth labels) corrected for chance. It ranges from -1 to $+1$, with $+1$ indicating

perfect agreement. ARI is particularly useful because it is invariant to label permutations (cluster 0 in prediction need not correspond to cluster 0 in ground truth).

c) *Normalized Mutual Information (NMI)*: This metric [30] measures the mutual information between two clusterings normalized by the entropy of each clustering. It ranges from 0 (no agreement) to 1 (perfect agreement).

III. SIMULATION RESULTS

A. Experimental Setup

All experiments were conducted on a system with 16 GB RAM and an 8-core processor. Text preprocessing included lowercasing, removal of punctuation and numbers, removal of English stop words, and stemming using the NLTK library. TF-IDF vectorization was performed with `max_features = 5000`, `min_df = 3`, and `max_df = 0.85` to control dimensionality and sparsity. For consistency, all K-means experiments used $k = 7$ clusters unless otherwise noted.

B. K-Means Clustering Experiments (Methods A–D)

Four K-means-based methods were evaluated to assess the impact of different sampling strategies on emotion clustering. Table II summarizes the experimental design.

All experiments were implemented in Python using the scikit-learn library. The general workflow follows a common pattern: (1) load and clean the dataset, (2) apply the specified sampling strategy, (3) vectorize the

TABLE II: Experimental Design for K-Means Methods

Method	Sampling Strategy	Sample Size	Purpose
A	Single random sampling	21,000	Baseline
B	Repeated sampling (10 iterations)	$10 \times 21,000$	Stability analysis
C	Stratified (natural distribution)	21,000	Preserve class imbalance
D	Balanced (3,000 per label)	21,000	Remove imbalance confounder

text using TF-IDF, (4) perform K-means clustering, and (5) evaluate the results using ARI, NMI, Silhouette, purity, and per-emotion recall.

1) *Method A: Single Random Sample*: Method A established the baseline using a single 21,000-instance random sample. The sample preserved the natural class distribution (joy: 30.2%, sadness: 25.6%, disgust: 4.2%, surprise: 4.8%). The optimal number of clusters determined by silhouette score was $k = 14$, substantially higher than the true number of emotion labels ($k = 7$).

TABLE III: Method A results

Metric	Value	Interpretation
ARI	-0.0056	Worse than random chance
NMI	0.0134	Near-zero mutual information
Silhouette	0.0084	Almost no cluster separation
Overall Purity	31.72%	Low cluster purity
Optimal K	14	Double true $k = 7$

The confusion matrix revealed that all clusters were dominated by either joy or sadness, with the largest cluster (10,324 samples, 49.2% of the data) containing 30.5% joy and 23.3% sadness. The maximum cluster purity was only 39.4%. As shown in Table IV, disgust achieved 85.01% recall, suggesting it has distinct lexical markers, while all other emotions had recall below 51%.

TABLE IV: Per-Emotion Recall: Method A

Emotion	Recall	Interpretation
Disgust	85.01%	Distinct linguistic patterns
Love	50.94%	Spread across clusters
Surprise	50.19%	Poorly captured
Joy	49.92%	Distributed across clusters
Fear	46.74%	Poorly captured
Anger	44.66%	Poorly captured
Sadness	44.59%	Poorly captured

a) Pseudocode Guide (method_A.py):

The implementation consists of several functions. `method_a_single_sample` draws a simple random sample using pandas function `sample()`. `create_cluster_label_confusion` constructs a confusion matrix showing the percentage of each cluster belonging to each true emotion label.

`evaluate_clustering` computes ARI, NMI, Silhouette, overall purity, and per-emotion recall. `run_method_a` runs the full experiment.

2) *Method B: Repeated Sampling*: Method B evaluated stability across 10 independent 21,000-instance samples. Two runs were conducted: (1) finding optimal

k across iterations, and (2) fixed $k = 7$ stability analysis.

TABLE V: Method B results (Fixed K=7)

Metric	Value	Interpretation
Mean ARI	0.0046 ± 0.0047	Near-random agreement
Mean NMI	0.0199 ± 0.0169	Very low mutual information
Mean Silhouette	0.0091 ± 0.0156	Almost no cluster separation
Centroid Similarity	0.0144 ± 0.0041	Very low stability
Pairwise ARI	0.0002 ± 0.0050	Assignments are random
Cluster Size CV	1.9762	Extreme size variation
Optimal K	12	Not 7

The optimal k determined by ARI maximization was 12, with a mean ARI of only 0.0083. For fixed $k = 7$, the mean ARI was 0.0046 ± 0.0047 , centroid similarity was extremely low (0.0144 ± 0.0041), and pairwise ARI between iterations was 0.0002 ± 0.0050 — indicating cluster assignments are near random. In each iteration, one cluster captured 90-99% of the data. The dominant cluster changed randomly across iterations, and negative silhouette scores in some iterations indicated the clustering was worse than random assignment. Fig. 1 shows the stability metrics and cluster size distributions for Method B in full.

a) Pseudocode Guide (method_B.py): The implementation performs stability analysis across multiple iterations. `get_sample_data` draws a random sample from the pre-vectorized full dataset and standardizes it.

`compute_cluster_metrics` computes ARI, NMI, and Silhouette for a single iteration. `run_optimal_k_analysis` runs K-means for K values 2 through 14 across 10 iterations to find the optimal K.

`run_fixed_k_stability` runs K-means with fixed $K = 7$ across 10 iterations.

`compute_stability_metrics` computes centroid similarity (via Hungarian assignment), cluster size coefficient of variation, and pairwise ARI. Plotting functions generate figures saved to `figures/`.

3) *Method C: Stratified Sampling*: Method C explicitly preserved the natural class distribution using stratified sampling. The sample perfectly matched the full dataset distribution, with minorities (disgust: 4.2%, surprise: 4.8%) severely under-represented.

Two clusters captured 72% of the data (Cluster 0: 50.1%, Cluster 4: 21.9%). Joy and sadness dominated every cluster. Disgust achieved 88.75% recall, while love had the lowest recall at 40.00%.

a) Pseudocode Guide (method_C.py): The implementation preserves natural class distribution.

TABLE VI: Method C results

Metric	Value	Interpretation
ARI	0.0040	Near-random agreement
NMI	0.0331	Very low mutual information
Silhouette	0.0086	Almost no cluster separation
Overall Purity	34.29%	Low cluster purity
Optimal K	14	Double true $k = 7$

`method_c_stratified_sampling` computes proportional sample sizes for each emotion label based on the full dataset distribution and samples accordingly.

`create_cluster_label_confusion` returns both percentage and count confusion matrices. `evaluate_clustering` computes metrics including cluster sizes. `run_method_c` runs the full experiment.

4) *Method D: Balanced Sampling*: Method D removed the class imbalance confounder by sampling exactly 3,000 instances from each emotion label. Three analyses evaluated scaling viability: centroid stability, learning curves, and out-of-sample validation.

The learning curve showed minimal improvement from 500 to 3,000 samples per label (+0.0074 ARI), indicating data quantity is not the bottleneck. The small train-test gap (0.0178) suggested the model generalizes, but to near-random predictions (ARI ≈ 0.02).

a) *Pseudocode Guide* (`method_D.py`): The implementation includes balanced sampling and three scalability tests. `method_d_balanced_sampling` draws exactly 3,000 instances from each emotion label.

`test_centroid_stability` runs K-means on five independent balanced samples using a fixed TF-IDF vectorizer and computes pairwise centroid distances. `test_learning_curves` tracks performance with increasing samples per label.

`test_out_of_sample_validation` computes train-test gap. `run_method_d` orchestrates all three tests and produces a scalability verdict.

C. Summary of K-Means Experiments

a) Consistent Findings Across K-Means Methods:

- 1) **All methods failed**: ARI remained near zero across all four methods (-0.0056 to 0.0231), indicating K-means on TF-IDF features cannot discover meaningful emotion clusters.
- 2) **Optimal K is not 7**: Methods A and C found $k = 14$, while Method B found $k = 12$, suggesting K-means detects lexical patterns rather than emotional categories.
- 3) **Cluster collapse**: All methods exhibited the "one large cluster" problem where 50-99% of data collapsed into one or two clusters.
- 4) **Disgust is separable**: Disgust consistently achieved 85-89% recall, suggesting it has unique lexical markers.

5) **Other emotions overlap**: Love, anger, fear, sadness, and surprise had recall below 52%, indicating high semantic overlap in TF-IDF space.

6) **Stability is poor**: Method B confirmed centroids are inconsistent, cluster assignments are effectively random, and cluster sizes vary wildly across iterations.

b) *Why K-Means + TF-IDF Fails*: The consistent failure can be attributed to three fundamental issues:

- 1) **TF-IDF cannot capture semantics**: TF-IDF treats words as independent features, failing to capture semantic similarity. Expressions like "I feel down" and "I'm feeling blue" convey the same emotion but share no TF-IDF overlap.
- 2) **K-means' spherical cluster assumption**: Emotion expressions form highly overlapping, non-spherical distributions. Silhouette scores near zero confirm clusters heavily overlap.
- 3) **Curse of dimensionality**: With 5,000 TF-IDF features, Euclidean distance becomes meaningless as all points become approximately equidistant [31].

c) *Implications for the Full Dataset*: The flat learning curve in Method D (ARI improvement of only 0.0074 from 500 to 3,000 samples per label) demonstrates that data quantity is not the bottleneck. Adding more data would not improve performance because the fundamental issue is the representation (TF-IDF) and algorithm (K-means), not data scarcity.

D. Alternative Approaches: Semantic Embeddings and Density-Based Clustering

To address the limitations identified in the K-means experiments, two alternative methods are introduced:

- **Method E: Word2Vec + K-means** — Replaces TF-IDF with Word2Vec sentence embeddings to test whether semantic representations improve clustering.
- **Method F: DBSCAN + TF-IDF + UMAP** — Replaces K-means with DBSCAN on UMAP-reduced TF-IDF features to test whether non-spherical clusters better capture emotion structure.

1) *Method E: Word2Vec + K-Means*: Method E represents each text as a 300-dimensional dense vector by averaging Word2Vec embeddings of its constituent words. This captures semantic similarity—for example, "I feel down" and "I'm feeling blue" have similar embeddings despite sharing no vocabulary.

Word2Vec produces more balanced cluster sizes (largest: 28.3%) compared to TF-IDF (49-50%), partially mitigating the "one large cluster" problem. However, the ARI (0.0199) remains near zero, and disgust recall drops from 85-89% to 55.03%, suggesting semantic averaging dilutes emotion-specific lexical markers. Love shows slight improvement (21.2% in one cluster), but overall performance remains near-random.

TABLE VII: Method D results

Metric	Value	Interpretation
ARI	0.0231	Near-random agreement
NMI	0.0515	Very low mutual information
Silhouette	0.0081	Almost no cluster separation
Centroid Stability	0.2432 $\pm$ 0.0234	Moderate instability
Train-Test Gap (ARI)	0.0178	Generalizes but poorly

TABLE VIII: Summary: All K-Means Methods

Metric	Method A	Method B	Method C	Method D
ARI	-0.0056	0.0046 $\pm$ 0.0047	0.0040	0.0231
NMI	0.0134	0.0199 $\pm$ 0.0169	0.0331	0.0515
Silhouette	0.0084	0.0091 $\pm$ 0.0156	0.0086	0.0081
Optimal K	14	12	14	7 (fixed)
Disgust Recall	85.01%	N/A	88.75%	N/A

TABLE IX: Method E results

Metric	Value	Interpretation
ARI	0.0199	Near-random agreement
NMI	0.0322	Very low mutual information
Silhouette	0.0047	Almost no cluster separation
Overall Purity	31.90%	Low cluster purity
Largest Cluster	28.3%	More balanced than TF-IDF
Disgust Recall	55.03%	Dropped from 85-89%

a) Pseudocode Guide (method_E.py):

The implementation uses Word2Vec embeddings instead of TF-IDF. `load_word2vec_model()` downloads the pre-trained Google News 300-dimensional model using `gensim.downloader.get_sentence_embedding(text, model, embedding_dim)` splits text into words, retrieves each word’s vector, and averages them to produce a sentence embedding (returns zero vector for empty texts). `run_method_e(df, sample_size, n_clusters, random_seed)` runs the full experiment.

2) Method F: DBSCAN + TF-IDF + UMAP:

Method F replaces K-means with DBSCAN, which does not assume spherical clusters. To address the curse of dimensionality, UMAP reduces the 5,000-dimensional TF-IDF space to 50 dimensions.

TABLE X: Method F results

Metric	Value	Interpretation
ARI	-0.0047	Worse than random chance
NMI	0.1334	Highest among all methods
Silhouette	-0.0622	Negative; poor separation
Overall Purity	87.12%	High but misleading
# Clusters	226	Extreme fragmentation
Noise Points	4,141 (19.7%)	Significant outliers

DBSCAN produces 226 clusters, far exceeding the true number of emotion labels. This fragmentation occurs for a number of reasons. The first is that DBSCAN finds dense regions wherever they exist. It is also due to high intra-class variation creating multiple clusters per emotion. Furthermore, overlapping emotions form

separate clusters, and finally because UMAP’s local structure preservation amplifies small differences.

Despite fragmentation, DBSCAN achieves the highest NMI (0.1334) and purity (87.12%) among all methods. Many clusters are 100% pure (e.g., 27 pure joy clusters, 32 pure sadness clusters). However, the ARI (-0.0047) and negative Silhouette (-0.0622) confirm poor overall clustering quality. Disgust achieves the highest recall (52.97%), consistent with earlier methods. Table XI shows the full list of cluster fragmentation.

TABLE XI: Cluster Fragmentation by Emotion (Method F)

Emotion	Pure Clusters ($\geq 90\%$)	Recall
Joy	27	42.95%
Sadness	32	38.31%
Fear	12	41.77%
Anger	10	38.74%
Love	8	44.98%
Surprise	3	44.50%
Disgust	0	52.97%

a) Pseudocode Guide (method_F.py): The implementation uses DBSCAN with UMAP dimensionality reduction. `run_method_f` runs the full experiment: sampling, TF-IDF vectorization, UMAP reduction to 50 components, feature scaling, DBSCAN parameter tuning via grid search over `eps` and `min_samples`, and evaluation.

`create_cluster_label_confusion` handles noise points (labeled ‘-1’) appropriately. `evaluate_clustering` computes ARI, NMI, and Silhouette (filtering out noise points), along with purity and per-emotion recall.

E. Summary of All Methods

Table XII shows an at-glance summary of all six methods. The obtained results can be summarized in a few points. Firstly, TF-IDF combined with K-means is unsuitable for emotion clustering, as all K-means methods (A-D) produced near-zero ARI scores, indicating the approach cannot distinguish between emotions. Furthermore, class imbalance is not the

TABLE XII: Comparison of all methods

Method	Algorithm	Features	ARI	NMI	Sil.	Purity
A	K-means	TF-IDF	-0.0056	0.0134	0.0084	31.72%
B	K-means	TF-IDF	0.0046	0.0199	0.0091	N/A
C	K-means	TF-IDF	0.0040	0.0331	0.0086	34.29%
D	K-means	TF-IDF	0.0231	0.0515	0.0081	N/A
E	K-means	Word2Vec	0.0199	0.0322	0.0047	31.90%
F	DBSCAN	TF-IDF+UMAP	-0.0047	0.1334	-0.0622	87.12%

primary issue; Methods A (random), C (stratified), and D (balanced) all produced similar results, confirming the failure is due to representation and algorithm rather than data distribution. While Word2Vec improves cluster balance (largest cluster: 28.3% vs. 49-50% for TF-IDF), it doesn't improve quality—disgust's separability was lost (85% $\rightarrow$ 55%). This suggests that semantic averaging dilutes emotion-specific markers. Moreover, DBSCAN captures structure but fragments emotions; it achieved the highest NMI (0.1334) and purity (87.12%) among all methods, but at the cost of extreme fragmentation (226 clusters), suggesting emotion expressions form many small, dense regions rather than well-separated groups. The fundamental challenge remains: regardless of representation (TF-IDF vs. Word2Vec) or algorithm (K-means vs. DBSCAN), the ground truth emotion labels don't correspond well to natural structures in the data, as emotions are expressed in highly diverse ways that do not form clean, separable clusters.

IV. CONCLUSION

A. Summary of Contributions

This paper systematically investigated unsupervised emotion clustering from imbalanced text data. Six approaches were evaluated: four K-means variants with different sampling strategies (single random, repeated, stratified, and balanced), K-means with Word2Vec embeddings, and DBSCAN on UMAP-reduced TF-IDF features. The main finding is that K-means on TF-IDF consistently fails to discover meaningful emotion clusters, with ARI values near zero across all sampling strategies. This failure stems from three fundamental issues: TF-IDF's inability to capture semantic similarity, K-means' spherical cluster assumption, and the curse of dimensionality. The Word2Vec-based approach produced more balanced clusters (largest: 28.3% vs. 49-50% for TF-IDF) but did not improve quality, and disgust's distinct separability was lost (85% $\rightarrow$ 55% recall). DBSCAN achieved the highest NMI (0.1334) and cluster purity (87.12%) among all methods, at the cost of extreme fragmentation into 226 clusters and a negative Silhouette score (-0.0622). Across all methods, emotion expressions form many small, dense regions rather than a few well-separated groups, and ground truth labels do not align with natural structures in the data.

B. Data and Code Availability

All the code referenced in this paper as well as the data, that are publicly available, can be found on Github: <https://github.com/kanitafro/emotions-data-k-means>

C. Limitations and Future Work

Several limitations should be acknowledged. The reliance on TF-IDF limits semantic understanding, and Word2Vec averaging loses word order and context. DBSCAN's fragmentation, while revealing the dense-region structure of emotions, produces too many clusters for practical use. Evaluation depends on ground truth labels that may be imperfect, as emotions are often ambiguous and multi-labeled. The dataset, while large, may not capture the full diversity of emotional expression across domains and cultures. Future work should explore transformer-based embeddings (BERT, SBERT) that capture context and word order. Hierarchical density-based clustering (HDBSCAN) could provide multi-granularity views, allowing users to choose the appropriate level of granularity. Soft clustering approaches such as Gaussian Mixture Models could better capture overlapping emotion distributions, and multi-label evaluation frameworks could better account for the inherent ambiguity of emotional expression.

D. Final Remarks

This paper demonstrates that unsupervised emotion clustering is fundamentally difficult due to the highly diverse and overlapping nature of emotional expression. Neither word-frequency representations nor semantic embeddings, neither centroid-based nor density-based algorithms, succeed in discovering clusters that cleanly correspond to predefined emotion categories. DBSCAN reveals that emotions form numerous small, dense regions rather than well-separated groups, suggesting that the traditional discrete emotion view may not reflect the true structure of emotional language. These findings highlight the need for more nuanced approaches to emotion understanding in NLP, moving beyond simple clustering toward models that capture the continuous, multi-faceted, and context-dependent nature of human emotion expression.

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